

Project 22: Movie Revenue Prediction

Introduction to Artificial Intelligence

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Preliminary Report

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1 Description of the Task

The primary objective of this project is to develop a machine learning model capable of predicting the global revenue of movies. Given a specific set of movie attributes such as budget, genres, original language, and release date. The model aims to output a precise financial prediction. We frame this project as a supervised regression task. Unlike a classification problem that would categorize movies into success levels, our goal is to predict a continuous numerical value that represent the total worldwide revenue in dollars. The challenge lies in the high variance of movie earnings and the need to capture non-linear relationships between production features and commercial success.

2 Description of the Dataset

The data for this project are sourced from Kaggle (*The Movies Dataset*). It provides a comprehensive set of metadata for several thousand films. From the available features, the following were selected as inputs to the model: `budget`, `genres`, `keywords`, `original_language`, `production_companies`, `production_countries`, `release_date`, `runtime`, `spoken_languages`, `cast`, and `director`. These span both numerical variables (e.g., `budget`, `runtime`) and categorical features (e.g., `genres`, `director`). Features such as `vote_average`, `vote_count`, and `popularity` were deliberately excluded, as they are only available after a film's release and would constitute data leakage. The target variable is the `revenue` column, representing the total global box office receipts.

3 Proposed Solutions & Algorithms

In this section, we describe the standard approaches for movie revenue prediction and select three models to implement and compare.

Linear Regression: This will serve as our baseline model. It is a simple and interpretable approach that assumes a linear relationship between input features (e.g., `budget`) and the target revenue. It allows us to evaluate the necessity of more complex architectures.

Random Forest: This ensemble learning method is widely used for its ability to handle non-linear dependencies and its robustness against outliers. By combining multiple decision trees, it provides a more accurate estimation than simple linear models.

Multilayer Perceptron (MLP): A Multilayer Perceptron is a fully connected feedforward neural network that learns non-linear relationships through successive layers of neurons and activation functions, trained via backpropagation. We include it to experimentally evaluate whether a neural network can compete with tree-based methods on tabular data, despite typically requiring more preprocessing and hyperparameter tuning.